Smart Nation Sensor Platform, allowing seamless coordination between transportation, environmental, and emergency agencies. This approach exemplifies how drones serve as *aerial data infrastructure*, not isolated tools, but active components of a connected urban decision-making network. Singapore's case demonstrates that when strategically governed and digitally synchronized, UAVs can significantly enhance *mobility efficiency, environmental planning, and sustainable city management*.

8.10.2 Case study: Dubai – urban air mobility (UAM) trials with passenger drone taxis

Dubai is recognized as one of the first cities in the world to actively test and demonstrate urban air mobility (UAM) through the use of autonomous passenger-carrying drones. The Dubai Roads and Transport Authority (RTA), in collaboration with international aerospace firms such as Volocopter and EHang, began flight trials of passenger drone taxis as early as 2017. These trials demonstrated the feasibility of short-range, point-to-point aerial mobility within dense urban environments, showcasing Dubai's commitment to devel oping futuristic and technology-driven transportation systems (Madi, et al. 2024).

The drones used in Dubai's UAM trials are typically electric vertical take-off and landing (eVTOL) aircraft equipped with redundant motors, autonomous navigation systems, and emergency backup protocols. These systems are designed to transport one or two passengers over short intra-city distances without the need for conventional runways. During the test flights, the aircraft operated from specially designed skyport stations, highlighting new possibilities for compact aerial mobility infrastructure integrated into urban space. These skyports are envisioned to link business districts, airports, tourism hubs, and healthcare facilities, significantly reducing travel time compared to road transport.